
paper_id: "PAPER_1123"
 title: "H₂O Maser Emission from J-Type Shocks: UQFF S(t) Compression Verification"
 session: 222
 date: "2026-04-19"
 author: "Daniel T. Murphy"
 status: production
 cvw: "v2.0.0"
 tags: [H2O maser, 22 GHz, J-type shock, water maser, collisional pumping, Ug1 jump]
 crosslinks: [PAPER_1121, PAPER_1122]
 sm_anchor: "CVW v2.0.0 - G6 SM Anchor Gate compliant"

PAPER_1123: H₂O Maser Emission from J-Type Shocks — UQFF S(t) Compression Verification

Abstract

Following arXiv:1306.5276 (2013), we implement a UQFF calculator for H₂O maser emission driven by J-type shocks. The 22.235 GHz water maser line ($6_{16} \rightarrow 5_{23}$ rotational transition) is collisionally pumped in post-shock gas at $T \sim 300$ -1000 K with densities $n \sim 10^5$ - 10^6 cm⁻³. This verifies the UQFF $S(t)$ compression model, where J-type shocks are interpreted as abrupt Ug1 jumps.

1. J-Shock Compression

The strong J-shock limit gives post-shock density:

$$n_{\text{post}} = 4 \cdot n_{\text{pre}}$$

and post-shock temperature:

$$T_{\text{post}} = \frac{3\mu m_p v_s^2}{16k_B}$$

2. Maser Pumping Window

Collisional inversion of the 22 GHz H₂O transition requires:

$$300 \text{ K} \leq T_{\text{post}} \leq 1000 \text{ K}$$

This constrains shock velocities to $v_s \sim 20$ -80 km/s. Below 300 K, collisional rates are insufficient; above 1000 K, collisional de-excitation quenches the inversion.

3. Maser Optical Depth

$$\tau_{\text{maser}} = \frac{n_{\text{post}} \cdot X(\text{H}_2\text{O}) \cdot h\nu \cdot l_{\text{gain}}}{4\pi\Delta v \cdot k_B T_{\text{post}}}$$

where l_{gain} is the coherent gain path length (~ 10 AU typical).

4. UQFF Interpretation

The J-type shock maps directly to an abrupt Ug1 jump in the UQFF framework: - $S(t)$ compression: density enhancement factor - The maser activation window validates the thermal structure of Ug1 discontinuities

Overall alignment: **80%** — datasets on shock velocities and maser luminosities verify the $S(t)$ compression model.

References

- arXiv:1306.5276 — H₂O masers from J-type shocks (2013).
- Elitzur, M. (1992). Astronomical Masers. Kluwer Academic Publishers.